

GENPEI ZHANG

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EDUCATION

University of Wisconsin–Madison

Ph.D. in Computer Science

Aug. 2026 – Present

Madison, WI, USA

University of Electronic Science and Technology of China

B.E. in Electronic Information Engineering

GPA: 3.65 / 4.00

Sep. 2021 – Jun. 2025

Chengdu, China

PUBLICATIONS

- [1] Genpei Zhang. MissPath-FM: Flow Matching with Structured Priors for Partially Observed Time Series. In *Advances in Neural Information Processing Systems (NeurIPS 2026)*.
- [2] Runmin Jiang*, Genpei Zhang*, et al. Simulating Cryo-EM: Cycle-Consistent Predictor-Corrector Diffusion with Biophysical Modeling. In *Transactions on Machine Learning Research (TMLR)*. [arXiv:2505.23444](#)
- [3] Xingzu Zhan, Runmin Jiang, Vatsal Gupta, Tanush Swaminathan, Yanwen Wang, Genpei Zhang, Haili Wang, Min Xu, et al. MicroFM: Physics-Guided Flow Matching for Isotropic Microscopy Reconstruction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026)*. [\[Paper\]](#)
- [4] Xingzu Zhan*, Genpei Zhang*, et al. T2M Mamba: Motion Periodicity-Saliency Coupling Approach for Stable Text-Driven Motion Generation. In *IEEE International Conference on Multimedia and Expo (ICME 2026)*. [arXiv:2602.01352](#)
- [5] Genpei Zhang*, Yuntian Yang*, et al. CryoDiff: Cryo-EM Synthesis via Biophysics and Cycle-Consistent Diffusion. In *CVPR 2026 Workshop on Synthetic Data for Computer Vision*.
- [6] Yijiang Li, Genpei Zhang, et al. EgoPrivacy: What Your First-Person Camera Says About You? In *International Conference on Machine Learning (ICML 2025)*. [arXiv:2506.12258](#)
- [7] Dazhou Yu, Genpei Zhang, Liang Zhao. PolyhedronNet: Representation Learning for Polyhedra with Surface-Attributed Graph. In *International Conference on Learning Representations (ICLR 2025)*. [arXiv:2502.01814](#)

* denotes equal contribution.

RESEARCH EXPERIENCE

Cryo-EM Data Synthesis with Cycle-Consistent Diffusion

Carnegie Mellon University, Pittsburgh, PA

Prof. Min Xu | Xu Lab | Computational Biology Department

Mar. 2025 – Jun. 2026

- Developed a physics-guided conditional cycle-consistent diffusion framework for generating cryo-EM micrographs with realistic biophysical variability and spatially adaptive noise;
- Integrated biophysical simulation and mask-aware contrastive learning to preserve structural consistency across synthetic and real cryo-EM domains;
- Evaluated synthetic data using FID/CMMD and downstream particle-picking and pose-estimation tasks, and developed an open-source simulation toolkit.

Egocentric Vision and Privacy Attack Research

University of California San Diego, Remote

Prof. Nuno Vasconcelos | Statistical Visual Computing Lab

Nov. 2024 – Jul. 2025

- Co-developed EgoPrivacy, an 819-wearer, 131-scene benchmark for measuring demographic, individual, and situational privacy risks in first-person video;
- Developed the Retrieval-Augmented Attack (RAA) pipeline that associates egocentric clips with third-person visual evidence to improve attribute inference;
- Conducted large-scale evaluations of vision-language models for privacy leakage in egocentric video.

Graph Learning and 3D Shape Representation Learning

Emory University, Remote

Prof. Liang Zhao | Department of Computer Science

May 2024 – Oct. 2024

- Developed PolyhedronNet, which represents 3D polyhedra as surface-attributed graphs and combines rigid transformation-invariant local encoding with hierarchical graph neural networks;
- Worked on data processing, experimental design, and result analysis across 3D shape representation benchmarks.

Deep Passport Ensemble for Multi-Model IP Protection (Undergraduate Thesis)

UESTC, Chengdu, China

Prof. Le Zhang | School of Communication and Information Engineering

Nov. 2024 – Jun. 2025

- Developed Co-Passport Ensemble (CPE), a system-level passport framework that distributes a master passport across multiple sub-models;
- Integrated trapdoor normalization, distributed validators, and dynamic regularization for multi-model intellectual-property protection.

SKILLS

Programming Languages: Python, C/C++, MATLAB, $\LaTeX$

Frameworks & Infrastructure: PyTorch, Hugging Face Transformers/Diffusers, CUDA, distributed training (DDP/FSDP), Git

Domain Tools: ChimeraX, CryoSPARC, RELION